



Recycled Absorptiometer: Converting E-Waste into Lab Instruments



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ABSTRACT

The purpose of this study is to address Egypt's Grand Challenges of pollution reduction, improved scientific environments, and waste recycling by developing a functional laboratory instrument built entirely from discarded electronic materials. Our solution is a portable absorbance meter (absorptiometer) designed to measure the concentration of dissolved substances in aqueous solutions using recycled LEDs, an LDR sensor, and a repurposed microcontroller. To achieve this, we designed, constructed, and tested a compact device that applies the Beer–Lambert Law to quantify absorbance through a 1 cm optical path. The prototype was engineered to meet design requirements, including sustainability, accuracy, portability, low cost, and full reliance on recycled components. Testing involved preparing standard solutions, recording absorbance values, and generating calibration curves to verify measurement reliability. Major findings through our analysis showed that the device achieved concentration measurements within $\pm 5\%$ error, demonstrating strong agreement with expected values and reliable repeatability across trials. The LED–LDR optical system performed stably inside a light-sealed chamber, and the recycled microcontroller successfully processed data and transmitted results via Bluetooth. In conclusion, this project proves that e-waste can be transformed into precise scientific equipment suitable for educational and field applications. The final prototype supports environmental sustainability while improving access to laboratory tools, showing that recycled electronics can meaningfully contribute to scientific and technological development.

INTRODUCTION

Egypt faces several pressing developmental challenges, many of which stem from growing environmental pressures and the rapid increase of electronic waste (e-waste). Among the Grand Challenges, four stand out as directly relevant to this project: the urgent need to reduce pollution in air, water, and soil; to adapt to the effects of climate change; to recycle waste for environmental and economic benefit; and to improve the scientific and technological environment for all.

The dramatic rise in global e-waste generation intensifies these interconnected issues. As shown in **Figure 1**, the amount of e-waste produced continues to increase while recycling rates remain critically low, resulting in significant environmental harm (Global E-Waste Monitor, 2024).

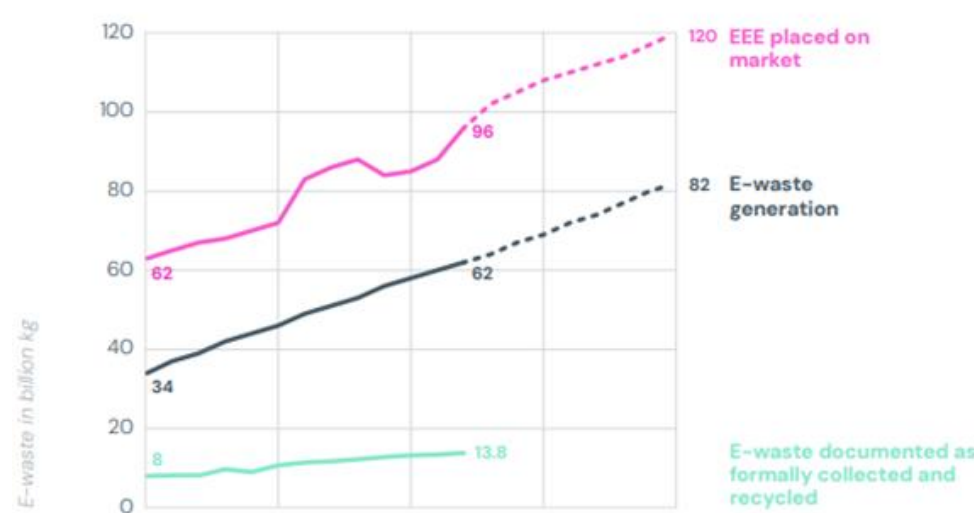


Figure 1: Graph showing the amount of E-waste produced and the volume recycled

Several attempts have been made to address Egypt's e-waste problem, each offering strengths but also revealing important limitations. Triple Re, Egypt's first certified WEEE recycler, has

processed over 5,500 tons of electronic waste and partnered with more than 200 organizations, demonstrating strong capacity and environmental safety. However, its overall impact is constrained by underutilized processing capacity, operating at only about 60%, due to limited raw material supply and insufficient storage space. On the other hand, the E-Tadweer application has made significant progress in public engagement by collecting thousands of tons of e-waste and incentivizing safe disposal through discounts and rewards. Yet, it also faces challenges, including limited drop-off locations, dependence on corporate partnerships, and reduced effectiveness when public awareness fluctuates. Both examples informed us of what works and what challenges persist in recycling E-waste.

To move beyond the limitations of prior solutions, our team identified key design requirements for a new solution. The device needed to repurpose electronic components directly from discarded devices, operate with high accuracy and environmental safety, remain compact and affordable for school laboratories' use, and serve as a practical scientific instrument that could contribute to improving Egypt's scientific and educational environment. These requirements guided the development of a solution that not only reduces waste but also promotes accessible learning.

With these considerations in mind, our team selected a portable absorbance meter constructed entirely from recycled electronic components. This device operates by emitting a narrow-band light beam through a liquid sample and measuring the transmitted intensity with an LDR sensor. Using the Beer–Lambert Law, the microcontroller calculates absorbance and converts it into concentration, allowing accurate measurement while being non-destructive to the liquid solution.

By developing a precise, eco-friendly scientific instrument from e-waste, our solution addresses Egypt's Grand Challenges while demonstrating how discarded electronics can be transformed into valuable educational tools to be used in laboratories.

The following section outlines the materials and methods used in constructing our prototype..

MATERIALS

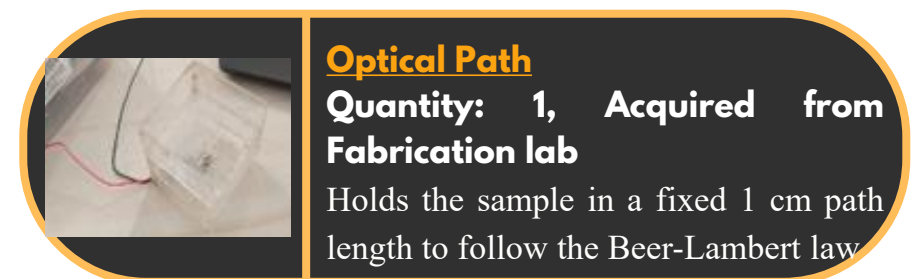
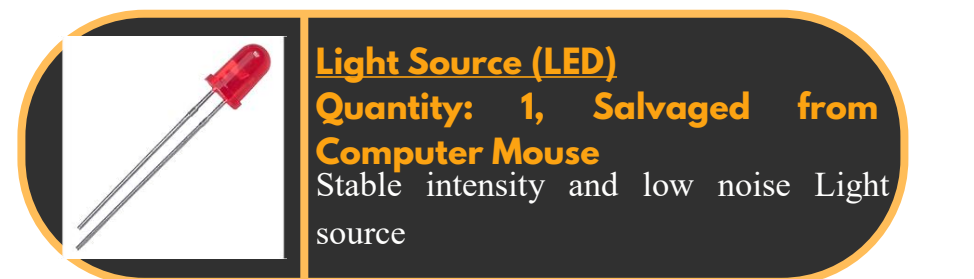
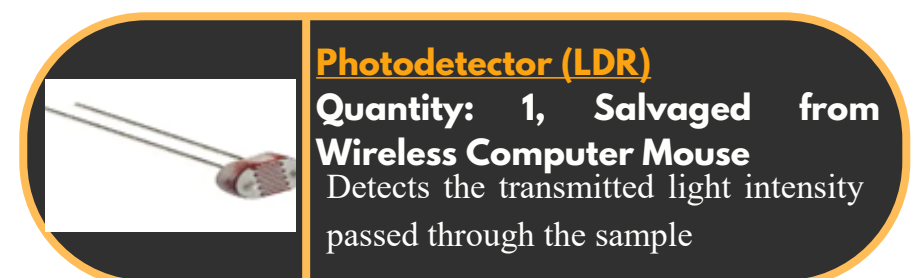
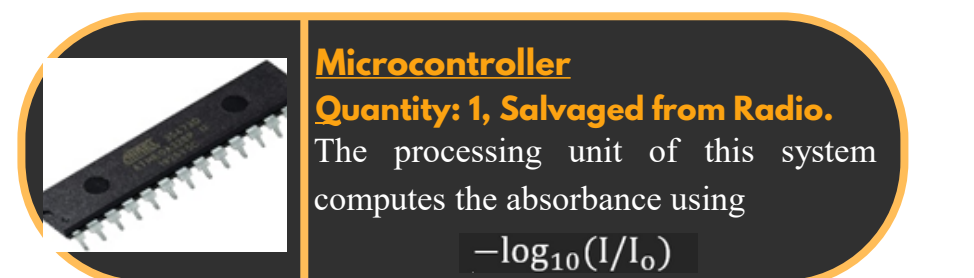
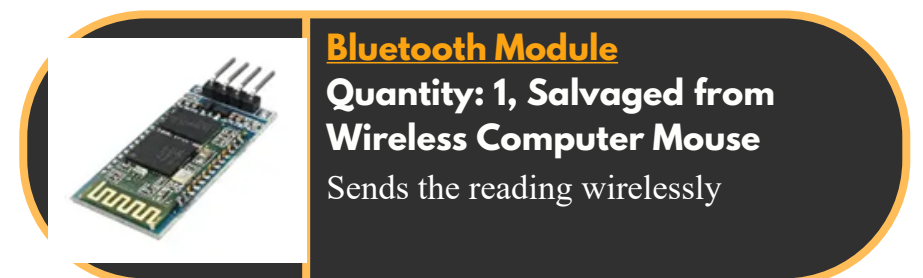
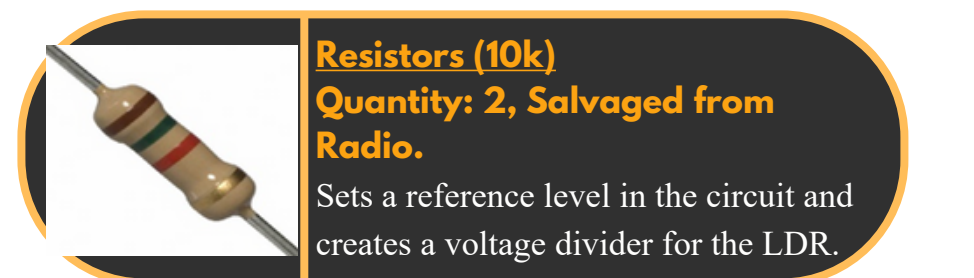
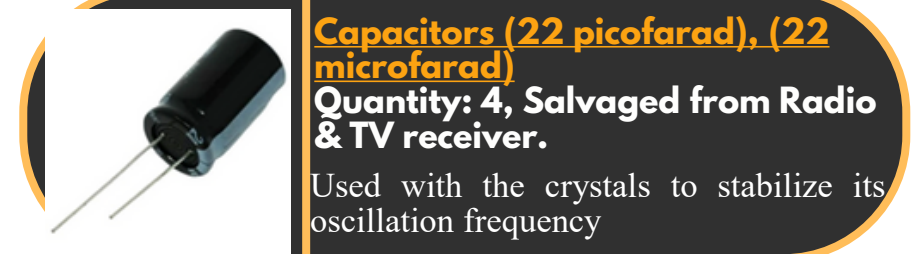
 Optical Path Quantity: 1, Acquired from Fabrication lab Holds the sample in a fixed 1 cm path length to follow the Beer-Lambert law	 Light Source (LED) Quantity: 1, Salvaged from Computer Mouse Stable intensity and low noise Light source
 Photodetector (LDR) Quantity: 1, Salvaged from Wireless Computer Mouse Detects the transmitted light intensity passed through the sample	 Microcontroller Quantity: 1, Salvaged from Radio. The processing unit of this system computes the absorbance using $-\log_{10}(I/I_0)$
 Bluetooth Module Quantity: 1, Salvaged from Wireless Computer Mouse Sends the reading wirelessly	 Resistors (10k) Quantity: 2, Salvaged from Radio. Sets a reference level in the circuit and creates a voltage divider for the LDR.
 Capacitors (22 picofarad), (22 microfarad) Quantity: 4, Salvaged from Radio & TV receiver. Used with the crystals to stabilize its oscillation frequency	 Voltage regulator Quantity: 1, Salvaged from a TV receiver. Supplies a stable and constant voltage

Table 1: Materials used in the construction of the prototype

METHODS

The prototype is made of 4 main steps:

1- Extraction and Bootloader Installation on the microcontroller

The process begins with removing the ATmega328P microchip from an old radio and preparing it for programming. To make the chip compatible with uploading code, a bootloader is burned onto it using the connection setup shown in **Figure 2**. This setup ensures that all necessary data pins are correctly linked for reliable data transfer to the ATmega328P.

2. Transferring Code to the microcontroller:

After installing the bootloader, the code is uploaded from the Arduino IDE to the microchip. This requires temporarily removing the microcontroller from an Arduino Uno board and connecting the serial communication pins directly to the standalone microcontroller, allowing the IDE to program it as an external device.

3. Fabrication of Acrylic Housing Components

Seven acrylic pieces were prepared in the fabrication laboratory to construct the device's housing: six sheets measuring 6×6 cm and one sheet measuring 6×2.5 cm. Two of the 6×6 pieces were further modified for specific functions, one to mount the LDR and another to mount the LED and allow solution discharge. These components form the structural enclosure of the final design.

4. Connection of the LDR in the Circuit

The LDR is integrated into the circuit by connecting one terminal to a 5V power supply, while the other terminal is routed into a voltage divider. This second terminal connects to a 10 kΩ resistor leading to ground, and the junction between the LDR and resistor is connected to the analog input pin A0. The full wiring layout is illustrated in **Figure 3**.

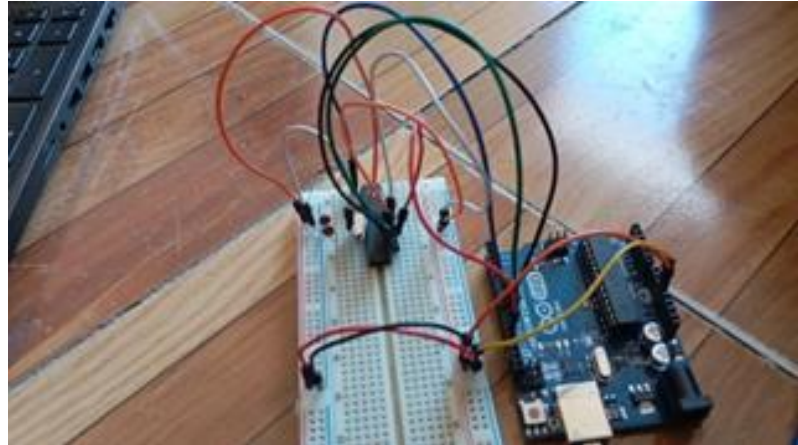


Figure 2 represents the wire connection for the boot loading process

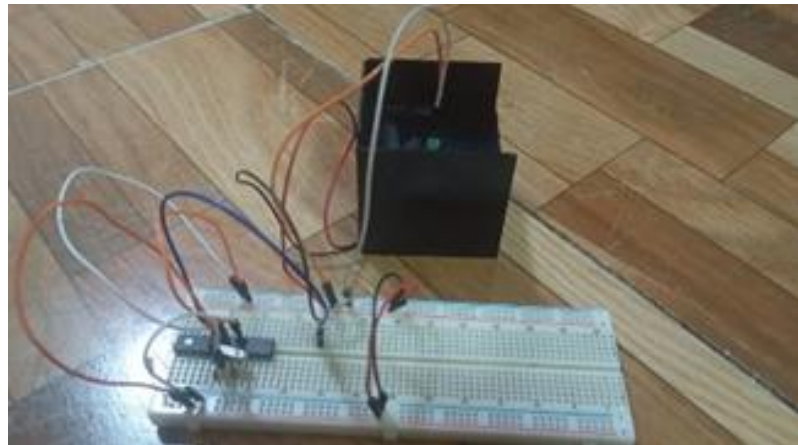


Figure 3 presents the connections of the LDR sensor and the LED

TEST PLAN

1. User-Friendly & Safety and Availability & readiness: Have 3 non-experts operate the device at any place and time using only basic instructions; all must succeed within 10 minutes and rate ease $\geq 4/5$.

2. Accuracy & Precision: Measure 3 known standards with 5 replicates each; confirm all measured concentrations have $\pm \leq 5\%$ error and consistent repeatability.

3. Volume Check: Measure prototype dimensions and confirm total volume $\leq 0.003 \text{ m}^3$.

RESULTS

Negative results:

Initial testing of the absorptiometer revealed some slight variation in the results. The prototype's accuracy was above $\pm 10\%$ and the results were unstable due to the light pollution coming from outside the prototype, since it was initially painted brown, which lets some light through it. Furthermore, there was an error in the code where the error percentage function was flawed, the voltage constant was wrong, and the selected height, leading to even worse results, as seen in **Table 2**. These challenges highlighted the need for improvement of the prototype.

Positive results:

After improving the painting of the housing and replacing the busted LDR and applying the fixes on the code, the accuracy of the prototype increased significantly, reaching $\pm 2.1\%$. One of the key improvements on the prototype is replacing the old LDR with another one, since the first one was flawed, improving the voltage readings, increasing the average readings from 4.0 V to 4.2 V, as seen in **Figure 4**.

The absorption reading improved from an average of 0.80 to an average of 1.192 due to the improvement of the LDR and the voltage readings. In addition, the concentration, which was measured using Lambert's law, improved due to its dependence on the absorption value, as seen in **table 3**.

As for the purity percentage after fixing the flawed law and using the correct one, The purity percentage measurement improved from 7% to 4.66%.

Parameters	value
Concentration	0.35
Absorption	0.80
Purity percentage	7%
voltage	4.0 V
height	1.2 cm

Table 2 represents the flawed results of the prototype before improvement

absorbance_A_sample: 1.192	A_water_ref: 0.050	Concentration(m/L): 0.350	Voltage: 4.200 V
absorbance_A_sample: 1.190	A_water_ref: 0.050	Concentration(m/L): 0.350	Voltage: 4.200 V
absorbance_A_sample: 1.190	A_water_ref: 0.050	Concentration(m/L): 0.350	Voltage: 4.200 V
absorbance_A_sample: 1.190	A_water_ref: 0.050	Concentration(m/L): 0.350	Voltage: 4.200 V
absorbance_A_sample: 1.190	A_water_ref: 0.050	Concentration(m/L): 0.350	Voltage: 4.200 V
absorbance_A_sample: 1.191	A_water_ref: 0.050	Concentration(m/L): 0.350	Voltage: 4.200 V
absorbance_A_sample: 1.191	A_water_ref: 0.050	Concentration(m/L): 0.350	Voltage: 4.200 V
absorbance_A_sample: 1.191	A_water_ref: 0.050	Concentration(m/L): 0.350	Voltage: 4.200 V

Figure 4: represents the improvement in the readings after the adjustments.

Trails	Concentrat ion (Molarity)	Absorbanc e	Purity percentage
Trail 1	0.392 M	1.090	4.50%
Trail 2	0.391 M	1.22	4.60%
Trail 3	0.390 M	1.10	4.70%
Trail 4	0.393 M	1.30	4.80%
Trail 5	0.392 M	1.15	4.60%
Trail 6	0.391 M	1.08	4.70%
Trail 7	0.394 M	1.25	4.90%
Trail 8	0.392 M	1.35	4.50%
Trail 9	0.393 M	1.18	4.60%
Trail 10	0.391 M	1.20	4.70%
Average	0.392 M	1.192	4.66%

Table 3 illustrates the concentration, absorbance, and the purity percentage after the improvements were made on the prototype



ANALYSIS

The prototype was tested, ensuring that it had met the minimum design requirement of measuring multiple measurements while having a measurement accuracy of no more than $\pm 5\%$ and having usage over various laboratories, while maintaining a simple and small design.

The first measurement is light absorption, which is measured by using the LDR that uses light intensity to change its variable resistance, where the higher the light intensity the less resistance and vice versa, by connecting one of the LDR's nodes to a power sources and connect the other node by a junction where it is connected to a voltage divider where a 10K ohm resistor is connected to the GND to increase the efficiency of the voltage readings, as seen in **Figure 5 (PH.1.04)** and one pin is connected to the analog input of the microcontroller, by doing this connections the microcontroller can analyze and the measure the output voltage resulted from the light intensity and then determining a voltage constant that changes according to the solute in the solution as seen in the following logarithmic curve (**MA.1.05**):

$$A = -\log_{10}\left(\frac{v}{v_0}\right)$$

$A = -\log_{10}\left(\frac{v}{v_0}\right)$ Where v represents the input voltage from the LDR and the v_0 represents the voltage constant, which was 63 according to the tested substance (copper sulphate). Determining the absorption, which is a unitless measurement. Absorption was measured to have an average value of $-\log\left(\frac{4.2}{63}\right) = 1.1$ across the conducted trials, as seen in **Figure 6**. This relation is derived from a former one that uses light intensity instead of voltage – since there is a direct relation between the light intensity and the net change in voltage therefore the logarithmic relation using voltage is derived. Concentration measurements were derived from the absorption measured by the microcontroller using Lambert's law, which utilizes absorption, concentration, and distance between the LDR and the LED, where absorbance and concentration and put in linear relation:

$$C = \frac{A}{\epsilon L}$$

Where the C represents the concentration and the ϵ represents the absorptivity constant, a constant that changes according to the solution used in the test case, which was the copper sulphate, which has an absorptivity constant of $2.81\text{M}^{-1}\cdot\text{cm}^{-1}$, and the L represents the distance between the LED and the LDR, which was 1cm. According to this relation, the average concentration across the trails was measured by $\frac{1.1}{2.81 \times 1} = 0.39\text{M}$, the average concentration value across each trail is put in comparison with the actual value as shown in **Figure 7**. The second measurement was the purity percentage, which is measured using the measured absorption value, which is the minimum absorption value representing the absorption of a clear solution, then they are put in a linear relation derived from Lambert's law:

$$\text{purity}(\%) = \frac{A_p}{A_s} \times 100$$

Where A_s represents the absorption of the sample, and the A_p represents the absorption of the clear sample, therefore the purity percentage is equal to $\frac{0.0539}{1.172} \times 100 = 4.6\%$. This law is the derivation of the law using concentration $\frac{C_s}{C_p} \times 100$.

then comes the calibration process where it helps in determining the accuracy of the measurement by comparing it to its actual real value. First, the calibration process of concentration, where the actual concentration of the solution used is measured and then compared to the output value measured by the instrument. To measure the concentration of the solution (Molarity), first, the number of moles of the solute must be measured by using its mass and molar mass of the solute **g/mol**, then determine the volume of the solution and divide the moles of solute by the volume of the solution (**CH.1.01**)

$$M = \frac{\text{moles of solute}}{\text{volume of solution in liters}} = \frac{6.4/160}{0.1} = 0.4\text{M},$$

after that we can use this law to measure our results accuracy

$$\left| \frac{\text{measured} - \text{True}}{\text{True}} \right| \times 100 = \left| \frac{0.3916 - 0.40}{0.40} \right| \times 100 = \pm 2.1\%$$

ensuring that it met the design requirement of an accuracy of $\pm 5\%$. Same with the purity percentage, where the actual value of purity percentage is measured with concentration by using actual concentration and the concentration reference that is equivalent to that of the absorption reference in value according to lambert's law the measurement accuracy of purity percentage

$= \left| \frac{4.5 - 4.4}{4.4} \right| \times 100 = \pm 2.1\%$, the accuracy of both concentration and the purity percentage is the same due to the linear relation between them where whenever one of them increases by a certain value the other increases by the same value (**MA.1.05**).

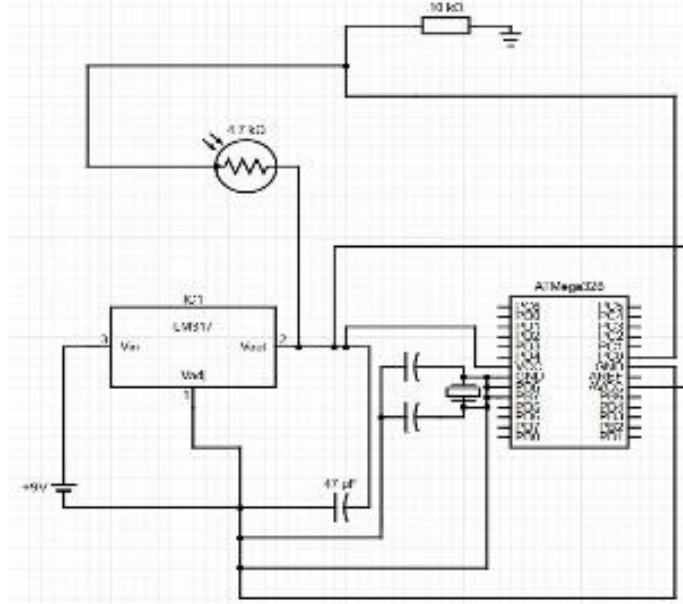


Figure 5: Schematic diagram prototype's circuit and connections.

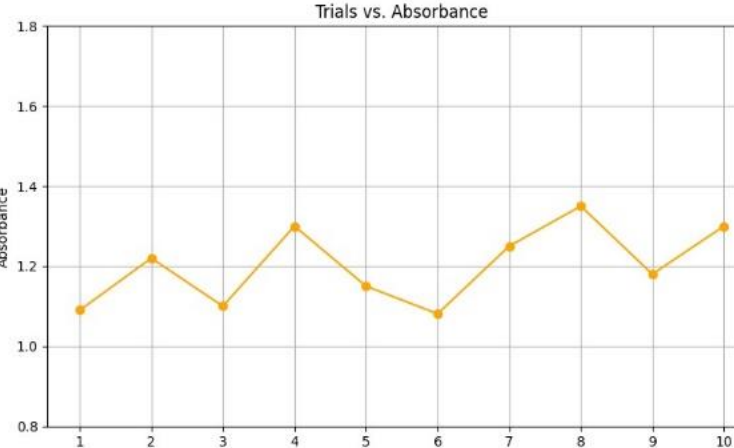


Figure 6: Graph showing the relationship between the average absorption of each trail and the actual concentration value.

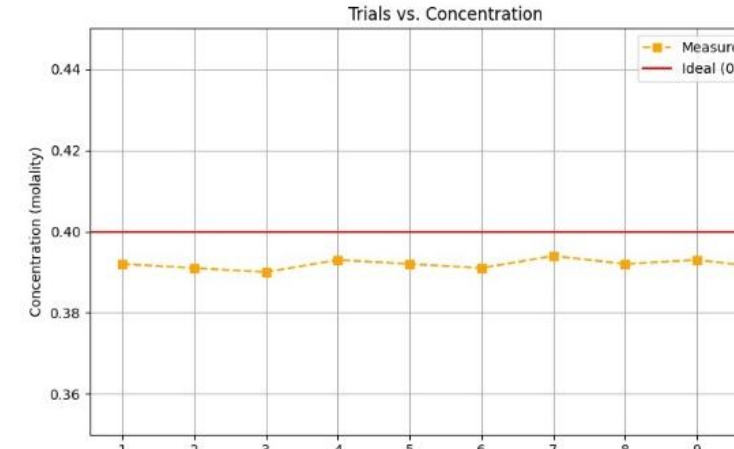


Figure 7: Graph represents the relationship between the average concentration (molarity) of each trail and the actual concentration value



CONCLUSION

The problem addressed in this project was the growing challenge of electronic waste in Egypt and the lack of affordable scientific instruments in schools and small laboratories. We aimed to create a functional absorptiometer built from recycled electronic components, offering both environmental benefit and practical scientific value. Through prototype development and testing, the device successfully measured solution concentrations with an error margin within $\pm 5\%$, confirming that repurposed circuitry, LEDs, and sensors can achieve reliable analytical performance when arranged in a controlled optical setup.

The results show that the prototype meets the intended design requirements: it is low-cost, constructed entirely from recovered materials, easy to operate, compact, and capable of generating consistent readings across multiple trials. These outcomes validate the central idea of the project: that meaningful laboratory tools can be developed from discarded electronics, reducing E-Waste, while developing education.

Compared to prior solutions, such as Triple Re's large-scale recycling operations or E-Tadweer's national collection system, our prototype serves a different but complementary purpose. Instead of managing waste at the industrial or governmental level, it demonstrates how small-scale, hands-on innovation can convert discarded technology into functional scientific equipment. This reduces waste but also increases the availability of analytical tools in resource-limited schools.

Thus, we could finally conclude through our analysis and results that the recycled components absorptiometer is a feasible and impactful solution with real potential for educational and low-cost laboratory applications.



RECOMMENDATIONS

The development and testing of the recycled component absorptiometer revealed several improvements that future teams could apply to enhance accuracy, durability, and real-world practicality. These recommendations aim to expand the device's functionality, enhance measurement stability, and support broader environmental applications, all while remaining aligned with the Grand Challenges of recycling, pollution reduction, and scientific accessibility.

The addition of Wavelength-Specific LED Modules for Multi-Analyte Detection:

A more advanced version of the device could include interchangeable LED modules (e.g., red, blue, green, and UV) to detect a wider range of chemical substances. Different analytes absorb at different wavelengths, and having multiple sources would allow the absorptiometer to test more pollutants and dyes.

Unfortunately, High-quality narrow-band LEDs for UV and blue wavelengths were impossible to come by in Discarded E-waste, which made them unavailable in our recycled components.

Integration of Automatic Temperature Compensation:

Because LED output and LDR resistance can shift with temperature, adding a simple temperature sensor (like a recycled thermistor) could correct absorbance readings automatically. This would improve reliability in outdoor applications such as water quality monitoring.

Development of a Mobile App for Data Logging:

Building a simple Android app would allow the device to transmit readings via Bluetooth into a GPS-tagged database. This will directly increase the Device's effectiveness and reliability.

Adding a Stable Regulated Power Supply:

Future versions of the absorptiometer could incorporate a voltage regulator to ensure the LED output remains constant, preventing measurement drift caused by battery fluctuations. Stable illumination dramatically improves accuracy and long-term reliability.

Recycled power regulation components, such as the LM7805, buck converter, were difficult to salvage in functioning condition, and new ones defeated the purpose of the project.



LITERATURE CITED

The Global E-waste Monitor 2024: Electronic waste rising five times faster than recycling — quantities, flows and the circular economy potential (GEM 2024). Global E-waste Monitor. https://ewastemonitor.info/wp-content/uploads/2024/03/GEM_2024_18-03_web_page_per_page_web.pdf. E-Waste Monitor Triple Re. (2025, September 5). *About Triple Re | first e-waste recycler in Egypt* <https://triple-re.net/en/e-waste-recycling/> World Bank. (2023, March 1). Egypt: New grant seeks to improve management of electronic and healthcare waste. <https://www.worldbank.org/en/news/press-release/2023/03/01/egypt-new-grant-seeks-to-improve-management-of-electronic-and-healthcare-waste> E-Tadweer app: Egyptians recycle electronic waste. (2021, September 23). Al-Ahram Weekly. <https://english.ahram.org.eg/NewsContent/50/1201/423528/AlAhram-Weekly/Egypt/E-Tadweer-app-Egyptians-recycle-electronic-waste.aspxapastyle.apa-2> CEDARE. (2024). E-Waste Management in Egypt: Status Report 2024. <https://2025.cedare.org/wp-content/uploads/2025/03/E-Waste-Management-in-Egypt-Status-Report-2024.pdf>



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